

Padlock Probes to Detect Single Nucleotide Polymorphisms

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Abstract

Rapid development of high-throughput DNA analyzation methods has enabled global characterization of genetic landscapes and aberrations in study subjects in a time and cost effective fashion. In most methods, however, spatial tissue context is lost since sample preparation requires isolation of nucleic acids out of their native environment. We hereby present the most recent protocol for multiplexed, in situ detection of mRNAs and single nucleotide polymorphisms using padlock probes and rolling circle amplification. We take advantage of a single nucleotide variant within conserved *ACTB* mRNA to successfully differentiate human and mice cocultured cells and apply presented protocol to genotype PCDH X and Y homologs in human brain. We provide a method for automated characterization and quantitation of target mRNA in single cells or chosen tissue area. mRNA of interest, harboring a polymorphism, is first reverse-transcribed to cDNA. Allele specific padlock probes are hybridized to the cDNA target and enzymatically circularized maintaining a physical link with the parent mRNA molecule. Lastly, circularized probes are replicated in situ, using rolling circle amplification mechanism to facilitate detection.

Key words Padlock probe, Single nucleotide polymorphism, SNP, mRNA genotyping, In situ, Single molecule, Single-cell

1 Introduction

Successful prevention of many diseases and accurate therapy guidance relies on the ability to correctly identify high-risk genetic aberrations that contribute to disease formation and progress. At present, most commonly practiced methods that allow for multiplexed mutation screening include primer extension [1, 2] and high-throughput new generation deep sequencing methods (NGS) [3]. Direct visualization of molecules, including mRNAs, overcomes the loss of information dictated by cell population averaging and provides perspective into spatiotemporal regulation of biological processes, often governed by single-cell, stochastic events. Recent developments of traditional fluorescent RNA in situ hybridization (FISH) including mRNA barcoding [4], different variants of in situ amplification like branched DNA (bDNA) FISH [5] and hybridization chain reaction (HCR) [6] have made

quantitation and localization of mRNAs feasible. Although various FISH approaches facilitate assessment of mRNA levels in individual cells, single nucleotide polymorphism (SNP) genotyping remains challenging since accurate sequence recognition is negatively correlated with length of probes hybridizing to the target mRNA.

This chapter introduces an up-to-date protocol and outlines the list of notes and hints to conduct successful multiplexed mRNA SNP detection in situ, using *padlock probes* (PLP) and *rolling circle amplification* (RCA). PLP is a continuous, single stranded DNA oligonucleotide with two terminal, target-specific arms (3' arm and 5' arm) connected by a DNA linker [7]. During target base pairing, PLP arms hybridize juxtaposed leading to probe circularization. Unlike traditional FISH approaches that solely rely on probe-mRNA hybridization, accuracy of SNP recognition with PLPs is driven by a mismatch-sensitive, thermostable DNA ligase. Differentiation between correct and incorrect target is practically binary and results in sealing the nick between probe arms rendering the probe circularized [8].

The PLP-based SNP detection protocol comprises series of enzymatic reactions, beginning with mRNA to cDNA reverse transcription (RT). To maximize the conversion rate, target specific RT primer contains several chemically modified DNA bases where 2'-O, 4'-methylene bridge locks the ribose pucker in C3' endo conformation [9]. Such "locked" nucleic acids DNA analogues (LNAs) display much higher RNA/DNA hybridization affinity than conventional nucleic acids. After the newly synthesized cDNA strand is chemically cross-linked with formaldehyde, original mRNA molecule is degraded by ribonuclease H (RNase H is an endoribonuclease that specifically targets and hydrolyzes RNA phosphodiester bonds in RNA/DNA heteroduplexes). Part of the mRNA that was hybridized to LNA bases within a RT primer is protected from degradation thus cDNA remains physically linked with the original target mRNA (Fig. 1). Next, allele-specific PLPs are hybridized and ligated to respective cDNA targets using *Thermus thermophilus* DNA ligase (*Tth* ligase) (*see Note 1*). Depending on the total length of hybridizing arms (typically between 30 and 40 bp) PLP makes ~2–4 full helical turns and becomes physically threaded around or "locked" to the cDNA. This stability allows for application of more stringent washing to ensure removal of incorrectly hybridized probes. Unidirectional rolling circle amplification of the target cDNA is initiated by the ϕ 29 DNA polymerase and templated by the target bound PLP [10]. DNA polymerase continuously extends the cDNA yielding 10^2 – 10^3 tandem copies of the original template (PLP) [11] maintaining physical contact with the original mRNA molecule (Fig. 1).

The single stranded RCA product (RCP) coils up spontaneously into ~500 nm size DNA blob which is "decorated" with fluorophore-conjugated oligonucleotides (detection/decorator

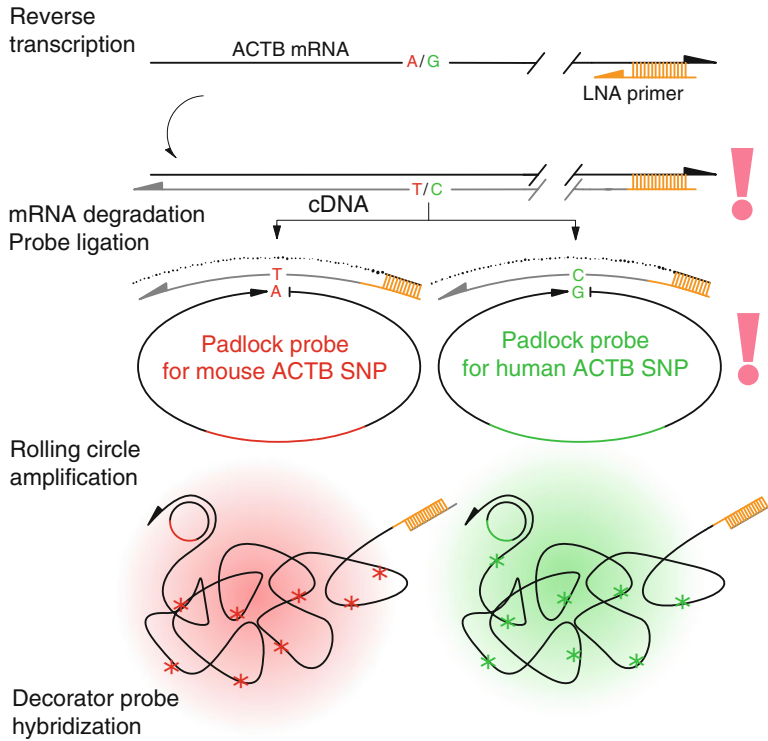


Fig. 1 Outline of in situ RNA genotyping using padlock probes and target primed rolling circle amplification. An mRNA molecule harboring a sequence variant (ACTB in the presented example) is converted into cDNA by reverse transcription, using target-specific LNA primer. Subsequently, the mRNA is degraded using RNaseH, padlock probes for the respective alleles hybridize to the target sequence and circularize upon DNA ligation. RCA generates a single-stranded DNA concatamer, which collapses into a typically half micrometer-sized DNA ball and contains hundreds of tandem repeated sequences that are complementary to the original padlock probe. Finally, fluorescently labeled decorator probes hybridize with their complementary motifs on the RCA product. Different DNA backbones of PLP for human and mouse allele allow efficient discrimination of RCP using unique, differently fluorophore-conjugated decorator probes (*green and red*). Exclamation mark: points in the protocol when the experiment can be paused (see **Note 23**)

oligonucleotides), complementary to the motif repeated in the amplicon. Local concentration of decorator probes creates a bright speckle readily differentiated from the background with standard EPI fluorescent microscope, even in samples with relatively high autofluorescence. By customizing PLP linker sequence, unique recognition motifs can be designed for each SNP or mRNA to be detected in the experiment. Multiplexing capacity is therefore limited to the number of decorator probes that can be differentiated with the available microscope system.

Each RCP represents a successful event of PLP hybridization, ligation, and amplification; thus, each observed signal represents a single SNP, making mRNA genotyping with padlock probes both qualitative and quantitative. Robustness and specificity of the protocol, combined with straightforward RCP localization allow user

for easy SNP and mRNA mapping with subcellular resolution. In the present chapter, we exploit a SNP within conserved *ACTB* mRNA to discriminate human and mice cocultured cells and apply described protocol to genotype *PCDH11* homologs in preserved brain tissue.

2 Materials

2.1 RT Primers

1. 100 μ M p_hsa_ACTB b 5'-C+TG+AC+CC+AT+GC+CC+ACCATCACGCCC.
2. 100 μ M p_mmu_ACTB b 5'-C+TG+AC+CC+AT+TC+CC+ACCATCACACCC.
+ = LNA-modified base.

For mRNA genotyping, we highly recommend using LNA modified primers [9] to maximize RT rate (random decamers can be used as universal RT primers, however the RT efficiency is lower probably due to lower hybridization strength). Primer should be designed in accordance with standard PCR primer design instructions (only a single, “reverse” primer is required for in situ RT). Primers should be 20–25 bp long, ~50% GC content and minimized probability of forming 3' hairpins, homodimer or heterodimer. Online tools like primer3 [12] are a great start as they suggest the best “right” RT primer with all thermodynamic parameters accounted for. LNA substitutions have to be suggested by the researcher and we typically substitute 3–7 DNA bases with LNA derivatives (*see Note 2*). Primer hybridization site should be downstream (3') and as close to the SNP as possible (*see Note 3* and Supplementary Note 3 in [13]). Once LNA primer sequence is determined, it should be evaluated for potential off target RT priming to prevent formation of false-positive cDNA and signals. Blastn against RefSeq mRNA database is recommended <http://blast.ncbi.nlm.nih.gov/Blast.cgi>. Figure 2 shows the *ACTB* mRNA in human and mouse with SNP highlighted, illustrating typical design example for both LNA primer and allele specific PLPs.

2.2 Padlock Probes and Decorator Probes

1. 2 μ M hsa_ACTB 5' GCCGGCTTCGCGGGCGA
AAAAAAAAAAAA *CCTCAATGCACATGTTTGGCTCC*
AAAAAAAAAAAA ACGGCGCCGGCATGTGCAAG.
2. 2 μ M mmu_ACTB 5' GCCGGCTTCGCGGGCGA
AAAAAAAAAAAA *CCTCAATGCTGCTGCTGTACTAC*
AAAAAAAAAAAA ACGGCGCCGGCATGTGCAAA.

Underline = target complementary arms, italic = detection probe complementary sequence.



Fig. 2 Padlock probe and LNA-primer design for human and mouse *ACTB* mRNA. **(a)** Pairwise alignment of mouse and human *ACTB* mRNAs with a SNP detected highlighted. **(b)** LNA primers (orange; LNA bases: **bold**; SNPs: green for human, red for mouse) hybridize with *ACTB* mRNA. During the ligation step, mRNA is degraded from mRNA/cDNA heteroduplex except where it was bound to LNA bases in RT primer. Padlock arms for matching allele hybridize with the target, while discriminative 3' base (symbolized as a triangle at the end of the 3' arm) is located over the interrogated T/C SNP. Reporter sequence (green/red) is amplified and used later for detection. **(c)** 5' → 3' full-length sequence of the padlock probe, with different parts of the probe indicated

Manual design of PLPs is possible and it can be done relatively quickly after gaining experience. However, to avoid making errors or for designing of multiple probes we recommend using open-source, automated ProbeMaker software [14] (see **Note 4**). PLP arms are designed to span SNP target in the target cDNA. For optimal and efficient SNP genotyping, we advise to adjust probe arms to that their melting temperatures (T_m) exceed ligation temperature by 5–10 °C considering ligation reaction conditions (see **Note 5**). If multiple probes are used in the same experiment, T_m of all PLPs arms should be similar to ensure comparable performance and minimize the background. As a rule of thumb, PLP backbone should be longer than total length of both arms (target) by approximately 10 bp (according to our experience, shortened backbones result in lower detection yield, probably because cDNA needs to “bend” to accommodate both PLP arms). Discriminating nucleotide of the PLP (hybridizing to the SNP) should always be placed on the 3' arm terminus [15] (in bold, see **Note 6**) and phosphate group on the 5' terminus to allow for probe ligation. PLP can be ordered pre-phosphorylated by a vendor of choice (pre 5'-phosphorylated oligonucleotides, synthesized by IDT (<http://idtdna.com>) using Ultramer[®] chemistry work well in our hands) or phosphorylated manually according to the standard phosphorylation protocol (see **Note 7**). After RCA, the “decorator motif” amplified in the RCP is utilized to visualize the amplification product through hybridization of a fluorescently labeled decorator oligonucleotides thus different motifs

are used for every target to allow discrimination (unless multiple SNPs are to be visualized simultaneously, *see Note 8*). Once the full PLP sequence is designed we strongly advice evaluation of probe secondary structure and off-target binding. We typically use mfold [16] (<http://mfold.rna.albany.edu>) or OligoAnalyzer for secondary structure predictions. Use prediction parameters adjusted for ligation reaction conditions (*see Note 5*). Every change in PLP sequence (different decorator motif or linker sequence) should be followed by a secondary structure prediction (*see Note 9*). To prevent formation of false-positive signals PLP target sequence (cDNA) should be blasted (<http://blast.ncbi.nlm.nih.gov/Blast.cgi>) against refseq mRNA database.

3. 10 μ M hsa_DO 5' CCTCAATGCACATGTTTGGCTCC.
4. 10 μ M mmu_DO 5' CCTCAATGCTGCTGCTGTACTAC.
Decorator probes (*see Note 8*).

2.3 Reagents

All enzymes should be stored at -20°C . Other reagents are stored at room temperature (RT) unless stated different.

1. 40 U/ μ L RNase Inhibitor (e.g., DNA Gdansk RIBOPROTECT).
2. TRANSCRIPTME—RnaseH-reverse transcriptase 200 U/ μ L and buffer (DNA Gdansk, *see Note 10*).
3. 40 U/ μ L Tth DNA ligase (e.g., GeneCraft).
4. 5 U/ μ L RNaseH.
5. 10 U/ μ L Phi29 DNA polymerase and buffer (e.g., Thermo Scientific).
6. Biological sample: cultured cells (alive) or tissue (fresh or fresh frozen). Fixed or freshly frozen tissues should be stored at -80°C .
7. Diethylpyrocarbonate (DEPC). Stored at 4°C (*see Note 11*)
Known carcinogen. Work under a fume hood, wear gloves.
8. RNase and DNase inactivators.
9. 70, 85, and 99.5 v/v % ethanol.
10. Xylene.
11. 0.1 M hydrochloric acid.
Highly corrosive. Work under fume hood with rubber PVC gloves and eyes protection.
12. 3.7% formaldehyde in PBS (*see Note 12*).
Known carcinogen. Use nitrile gloves, eyes protection and handle powder in the chemical fume hood.
13. Pepsin, lyophilized powder, 2.500 U/mg protein (*see Note 13*).
14. 0.25%trypsin–EDTA.

15. SlowFade[®] Gold Antifade Mountant (Thermo Scientific) or equivalent mounting medium (*see* **Note 14**).
16. 100 mM Hoechst 33342.
Store stock Hoechst 33342 solution at -20°C . Working solutions (protected from light) can be stored at 4°C for a couple of months.

2.4 Buffers and Solutions

All concentrated buffers are provided together with enzymes and stored according to vendor specification. Custom-made buffers should be prepared from DEPC-treated PBS or ddH₂O (*see* **Note 15**) and can be kept at RT for up to one month. We also recommend aliquoting DEPC-ddH₂O and DEPC-PBS into smaller volumes (50 mL) to minimize contamination.

1. Phosphate buffered saline (1× PBS): 137 mM NaCl, 10 mM sodium phosphate, 2.7 mM KCl, and DEPC-ddH₂O pH 7.4.
2. Washing buffer (PBS-T): 0.05% Tween 20 in 1× PBS.
3. Saline-sodium citrate buffer (20× SSC): 3 M NaCl and 300 mM trisodium citrate in DEPC-ddH₂O pH 7.
4. RT-mix (50 μL per chamber): 5 μL 10× RT-reverse transcriptase buffer, 1 μL 40 U/ μL RNase Inhibitor, 0.5 μL 20 $\mu\text{g}/\mu\text{L}$ BSA, 2.5 μL 10 mM dNTPs mix, 0.5 μL 100 μM LNA primer or 2.5 μL 100 μM random decamers, and TRANSCRIPTME-reverse transcriptase (we typically use 5 U/ μL for cell lines and 20 U/ μL for tissue sections). Fill up to 50 μL with DEPC-ddH₂O.
5. Ligation mix (50 μL per chamber): 5 μL 10× Tth ligase buffer, 2.5 μL 2 μM padlock probe(s), 4 μL 5 U/ μL RNase H, 0.5 μL 20 $\mu\text{g}/\mu\text{L}$ BSA, 2.5 μL 1 M KCl, 10 μL 100% formamide, and 1.25 μL 200 U/ μL Tth ligase. Fill up to 50 μL with DEPC-ddH₂O.
6. RCA mix (50 μL per chamber): 5 μL 10× Phi29 DNA polymerase buffer, 1.25 μL 10 mM dNTPs mix, 0.5 μL 20 $\mu\text{g}/\mu\text{L}$ BSA, 5 μL 50% glycerol, and 5 μL 10 U/ μL Phi29 DNA polymerase. Fill up to 50 μL with DEPC-ddH₂O.
7. Hybridization mix (50 μL per chamber): 5 μL 20× SSC, 10 μL 100% formamide, 0.5 μL 10 μM Decorator probe(s), 1.5 μL , and 100 mM Hoechst 33342. Fill up to 50 μL with DEPC-ddH₂O.

2.5 Equipment

1. Diamond pen.
2. Forceps.
3. 37°C incubator.
4. 45°C incubator.

5. Humidity chamber (e.g., empty slide or tips box with paper towel soaked with water).
6. Fluorescence microscope.
7. Secure-Seal hybridization chambers (Grace Bio-Labs) (*see Note 16*).
8. Coverslips (*see Note 17*).
9. Coplin jars.
10. Adhesive microscopy slides (e.g., Menzel Gläser SuperFrost®).
11. 150 mm × 25 mm culture dish used for cell fixation.

3 Methods

3.1 General Guidelines and Controls

RNases can maintain enzymatic activity even after prolonged autoclaving. Therefore, special measures should be taken when working with RNA.

1. We recommend designating bench area dedicated to RNA work only.
2. All bench surfaces, pipettes, or glassware should be treated with commercially available RNase- and DNase-inactivating agents. We usually wipe benches with 100% ethanol after such treatment.
3. Sterile, disposable, free plasticware (pipette tips, slide boxes, tubes, and flasks) work best in our hands and ensure RNase-free conditions.
4. We recommend validating efficiency and specificity of new PLPs on synthetic DNA targets.
5. Ligation fidelity can be monitored in vitro as a high molecular weight band on denaturing PAGE gel (linear PLPs and templates migrate faster than circularized PLPs).
6. RCA can be monitored in vitro (templates provide the 3' –OH group as a RCA primer, just as cDNA in regular protocol) by staining RCPs with either intercalating dyes (SYBR dyes) or decorator probes and visualized under a microscope (5–10 µL of stained RCA mix can be mounted on a microscope slide) or q-PCR system.
7. As a biological positive and negative controls, cell lines with/without target of interest provide a good model to evaluate assay specificity and sensitivity.
8. In accordance with good research practice, different day replicates are recommended, since variation in handling slides or in cell lines growth may influence the final result.

9. Finally, we recommend frequent assessment of the whole genotyping procedure on well-established system (*see Note 18*) to ensure maximum detection sensitivity.

3.2 Specimen Pretreatment

3.2.1 Adherent Cells

1. Culture cells in a flask until 80–90% confluence.
2. Wash cells twice with one PBS, add dislodge cells by adding 1 mL/25 cm² 0.25% trypsin–EDTA for a 1–2 min. Aspirate Trypsin–EDTA and monitor cells trypsinization under a bright-field microscope.
3. Resuspend the cells in appropriate culturing medium (FBS in a medium inactivates the trypsin).
4. Place sterile microscope slides in a 150 mm × 25 mm culture dish and add ~22 mL of medium to cover the slides. Cells adhere to most type of slides.
5. Carefully, seed total 3 mL of suspended cells directly on the slides (*see Note 19*).
6. Carefully, transfer the closed dish into the incubator and incubate the cells under appropriate conditions to allow them to adhere to the microscope slides (*see Note 20*).
7. Wash the slides twice with cold 1× PBS and transfer the slides to a sterile Coplin jar or sterile slides transport box.
8. Fix the cells with 3.7% formaldehyde at room temperature for 20 min.
9. Discard the formaldehyde and wash the slides twice with 1× PBS (*see Note 21*).
10. Gradually dehydrate and pre-permeabilize the cells in ethanol series (70, 85, and 99.5% each step for 3 min).
11. Air-dry the slides and store at –80 °C (long-term storage) or –20 °C (up to 2 weeks).
12. Use pencil to note a fixation date, cell type and number of the passage on the slide's writable area (*permanent markers may dissolve upon contact with ethanol and cause severe autofluorescence*).
13. Upon use, thaw slides at room temperature.
14. Attach Secure-Seal chamber(s) and rehydrate the cells by adding 1× PBS-T into the chamber (*see Note 22*).
15. Remove DEPC-PBS-T and permeabilize the cells by adding 0.1 M HCl for 5 min.
16. Remove HCl and wash the cells twice with PBS-T (*see Note 23*).

3.2.2 Freshly Frozen Tissue

1. Tissue sections, mounted on microscope slides (*see Note 24*) are stored at –80 °C until use.

2. Thaw slides at RT.
3. Fix the tissue in $1 \times$ PBS for 45 min. Use a sterile Coplin Jar or perform fixation directly in the chamber (depending on specimen area).
4. Wash once with $1 \times$ PBS for 5 min.
5. (Optional) Permeabilize the tissue by incubating with pre-warmed pepsin in 0.1 M HCl at 37 °C. Digestion with 250 U/mL (0.1 mg/mL) pepsin for 5 min is a good starting point in our experience. Optimal conditions need to be identified for the respective specimen (*see* **Notes 13** and **25**).
6. Wash once with $1 \times$ PBS for 5 min.
7. Dehydrate the tissue section in the ethanol series (70, 85, and 99.5% ethanol, 3 min each).
8. Air-dry and mount Secure-Seal chambers (if fixation was conducted in the Coplin Jar).
9. Rehydrate the tissue by in $1 \times$ PBS-T.

3.2.3 Formalin-Fixed, Paraffin-Embedded (FFPE) Tissue

1. Tissue sections, mounted on microscope slides (*see* **Note 24**) are stored at -80 °C until use.
2. Thaw samples at RT.
3. Dewax samples by submerging slides in a solvent series in separate Coplin jars:
15 min xylene, 10 min xylene, 2×2 min 100% ethanol, 2×2 min 95% ethanol, 2×2 min 70% ethanol, 5 min DEPC- H_2O , 5 min $1 \times$ PBS.
4. Permeabilize the tissue by incubating with pre-warmed pepsin in 0.1 M HCl at 37 °C (*see* **Notes 13** and **25**).
5. Wash with $1 \times$ PBS for 5 min.
6. Postfix the tissue in 3.7% formaldehyde for 10 min at room temperature.
7. Wash twice with $1 \times$ PBS for 5 min.
8. Dehydrate the tissue section in the ethanol series (70, 85, and 99.5% ethanol, each step for 3 min).
9. Air-dry the slides and mount Secure-Seal chambers.
10. Rehydrate the tissue with $1 \times$ PBS-T.

3.3 SNP Detection Protocol

Below, we present an updated protocol for mRNA SNP genotyping in cell lines, fresh frozen and FFPE tissue sections. We operate at 50 μ L reaction volumes that can be adjusted if necessary (use larger Secure-Seal chambers, *see* **Note 16**). To minimize introduction of contaminants into chambers, work in as clean environment as possible—place a clean paper towel between lab bench and the slide; change gloves and washing buffers frequently. At

temperatures above room temperature, Secure-Seal chamber inlets should be covered with standard PCR film to prevent evaporation of reaction mix. To further reduce evaporation, all incubations and reactions are performed in a humidified box.

3.3.1 Reverse Transcription

1. Apply the RT mix to the chamber.
2. Cover the chamber inlets and incubate the slides at 45 °C. The optimal incubation time needs to be determined experimentally. One hour long RT when using LNA-modified target-specific primers is usually sufficient. RT with random decamers is typically conducted overnight.
3. Wash the chamber once with 1 × PBS-T.

3.3.2 Postfixation

Postfixation is an important step during which the newly synthesized cDNA strand cross-links to the adjacent chemical groups of proteins side chains.

1. We typically fix cell culture for 10 min and tissue sections for up to 45 min at room temperature in 3.7% formaldehyde (*see Note 12*). Postfixation time should also be optimized for every specimen.
2. Wash the chamber twice with 1 × PBS-T.
3. At this point, the protocol can be halted and samples stored for a couple of days at 4 °C in 1 × PBS.

3.3.3 mRNA Degradation, Padlock Probe Hybridization and Ligation

1. Apply the Ligation mix to the chamber.
2. Cover the chamber inlets and incubate the slides at 37 °C for 30 min.
3. Transfer the slide to 45 °C incubator for 45 min (*see Note 26*).
4. Wash the chamber twice with 1 × PBS-T.
5. At this point, the protocol can be halted and samples stored for a couple of days at 4 °C in 1 × PBS.

3.3.4 Rolling Circle Amplification

1. Apply the RCA mix into the chamber.
2. Cover the chamber inlets and incubate the slides at 37 °C for ≥1 h (*see Note 27*).
3. Wash the chamber twice with 1 × PBS-T.
4. At this point, the protocol can be halted and samples stored for a couple of days at 4 °C in 1 × PBS.

3.3.5 Decorator Probe Hybridization and Nuclei Counterstaining

From this step onward, light-sensitive compounds will be used. Protect decorator probes as well as DNA intercalating dyes from prolonged, direct light exposure. Samples should be kept in a dark (covered) during and after incubations.

1. Apply the Hybridization mix to the chamber.
2. Incubate the slide at RT for ~20 min (*see Note 28*).
3. Wash the chamber twice with $1 \times$ PBS-T.
4. Mark the position of the chamber on the backside of the slide with a diamond pen and remove the Secure-Seal chamber (*see Note 29*).
5. Passing the specimen through an ethanol series (70, 85, and 99.5% ethanol, each for 3 min) dehydrates sample, cleans the slide from contaminants and removes any glue residues.
6. Air-dry, apply Slow-Fade medium on the coverslip (or specimen directly) and gently, mount the specimen (*see Note 30*).
7. Use pencil to note date and type of experiment on the slide's writable area (*permanent markers may dissolve upon contact with ethanol*).
8. Signal and specimen are stable for a long time (blue-yellow visible light spectrum dyes for at least one year, far red dyes are less stable) when kept at 4 °C and protected from light.

3.3.6 Image Acquisition and Analysis

1. Mounted slides can be imaged immediately. Conventional wide-field epifluorescence microscopes are usually sufficient to image RCA products in tissue sections and cells.
2. Make sure to use filters appropriate for nuclear staining and fluorophores used in the experiment (*see Note 31*).
3. Depending on the level of detail desired, use appropriate objective (we typically use $10\times$, $20\times$ and $40\times$ high numerical aperture objectives).
4. To ensure accurate signal quantitation during image analysis, carefully adjust exposure time for a signal channels (abundant RCPs might be hard to segment if overexposed).
5. We typically acquire 5–10 μm thick z-stack of images to capture RCPs in all focal planes (if possible, preview the specimen to define "first" and "last" stack corresponding to RCPs present in the lowest and highest stack and capture all images within this range).
6. Z-Stacks with multiple focal planes are combined to a single 2D maximum intensity projection (MIP) image that can be used in image analysis. We recommend taking images from several positions for each experiment (depending on cell density) to account for cell-to-cell expression variations.
7. Open-source programs such as CellProfiler [17] or ImageJ can be used for image analysis (*see Note 32*).

3.4 Anticipated Results

1. Every discrete, fluorescent detected signal originates from a single, successful detection of the SNP (Fig. 3).

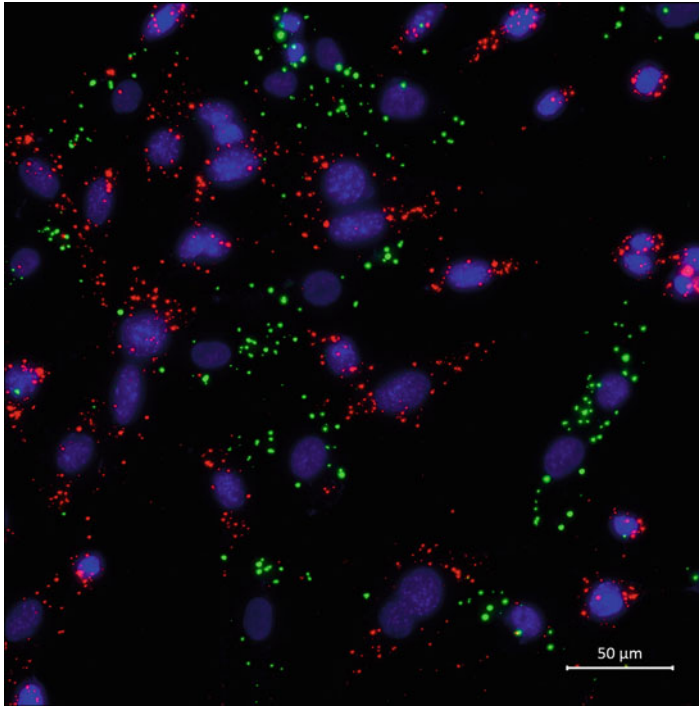


Fig. 3 In situ SNP detection of *ACTB* mRNA in human BjhTERT and mouse MEF cell line. Detection of *ACTB* SNP in human and mouse cell lines is represented as color-coded RCPs. *Green*: Detection of human *ACTB*; *Red*: Detection of mouse *ACTB* SNP. Size of the scale bar in the *inset* 50 μm

2. All RCPs should be localized to the cytoplasm, however RCPs can occasionally dissociate out of cells and generate RCPs on the glass slide (such signals should be disregarded).
3. RCPs can be easily distinguished from cellular autofluorescence by their defined brightness, shape and size.
4. Finally, RCP signal should be visible only in a single, fixed fluorescence channels (checking other channels is a good measure to evaluate signal specificity). Figure 3 shows *ACTB* mRNA genotyping in human BJhTERT and mouse MEF fibroblast cells. Mutation within the gene allowed designing two PLPs, which arms differ by only a single, 3' terminal base (G/A in human/mouse). When PLPs for both alleles were used together in cocultured cells, SNP specific signals were exclusively generated in either human or mouse cells.
5. We applied the protocol to genotype and differentiate *PCDH* X and Y chromosome homologs in human embryo medulla oblongata section. This study provided detailed insight into the spatial distribution of *PCDH11X/Y* and *NLGN4X/Y* in human developing nervous tissue. Observed patterns suggested the significant development complexity in the male central nervous system even if single cell expression data was

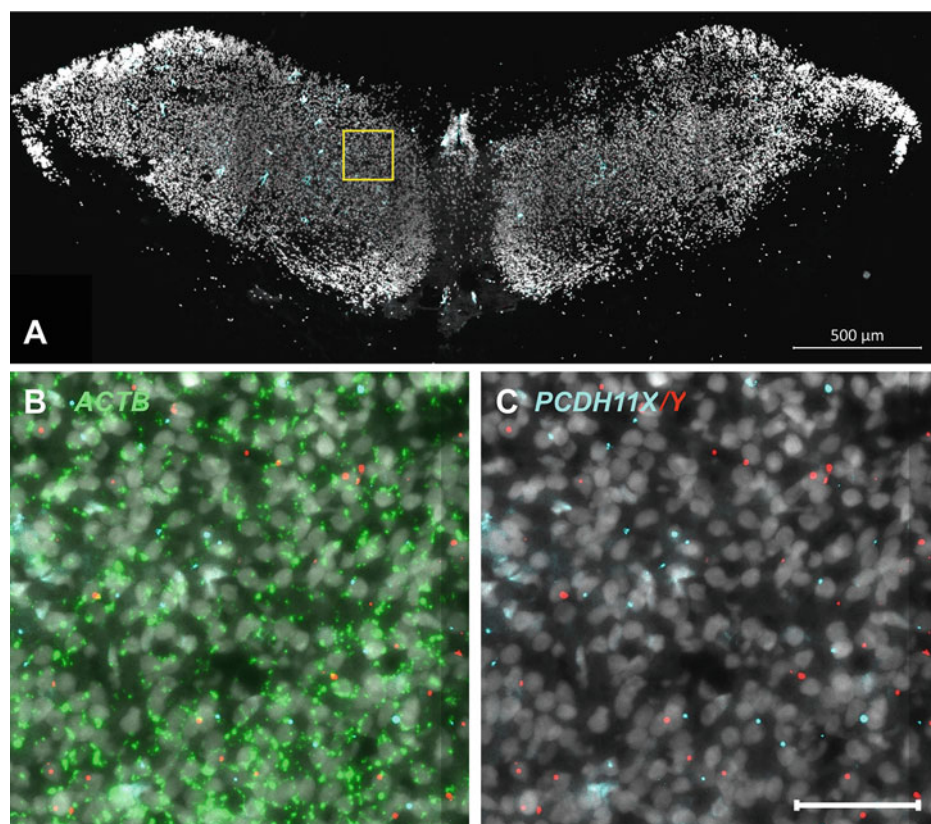


Fig. 4 In situ *PCDH11X/Y* SNP detection in male, Medulla oblongata embryo. Human male Medulla oblongata section is simultaneously hybridized with a single *ACTB* PLP, five probes for X chromosome *PCDH11* homolog and five for Y chromosome homolog of *PCDH11*. (a) Full Medulla oblongata section scan. *Inset* highlighted in yellow enlarged in **b**, **c**. (b) The same region of the section with staining for *ACTB* (green) and (c) *PCDH11X/Y* (cyan and red respectively). Size of the scale bar in the *inset* 50 μm. Probe sequences and experiment setup published in [18]

limited (*see* **Note 33**). Figure 4 shows a selected example of *PCDH X/Y* genotyping. Whole study including technical aspects and results is published in [18].

4 Notes

1. *Tth* DNA ligase exhibits significantly greater thermal stability and ligation specificity over traditional DNA ligases [15] and it works very well for SNP genotyping in our hands. Greater polymorphism discrimination specificity is generally observed when the discriminating nucleotide is present on the 3' end of the PLP arm [15]. Alternative DNA ligases can be used for conventional mRNA detection and genotyping [13] or

mitochondrial DNA genotyping [19], however the SNP detection specificity should be evaluated experimentally for every target with appropriate controls (*see* Subheading 3.1).

2. LNA modified primers can be purchased from EXIQON (<http://www.exiqon.com>) with standard desalting as an adequate purification method. Use TE buffer (pH 8.0) to resuspend oligonucleotides to 100 μ M stock. Accurate LNA bases substitution is crucial for optimal primer functionality. Too few LNA substitutions will have a minimal impact of the primer melting point, while too many may stabilize any predicted hairpins, homodimers, or off-target binding. Once primer sequence is known, we strongly advice computing primer secondary structure using any web-based secondary structure prediction tools. The OligoAnalyzer 3.1 tool from Integrated DNA Technologies (IDT), <http://eu.idtdna.com/calc/analyser> works good in our hands as many parameters that usually influences DNA folding can be specified (RT conditions presented in this chapter are the following: LNA primer concentration 1 μ M; monovalent and divalent cation concentration 50 mM and 4 mM respectively; and RT temperature 45 $^{\circ}$ C). LNA bases should be introduced as close to the primer 5' end as possible, interspaced with one or more conventional DNA bases. Consecutive LNAs, evenly distributed in the primer sequence or 3' end concentrated LNA substitutions should be avoided [20]. Based on secondary structure and homodimer formation predictions, LNAs should not be introduced in regions where dimer or hairpin formation probability is high. If no such positions are available, another primer from the primer3 list can be evaluated.
3. In our experience, longer distance between the SNP and the primer hybridization site negatively influences successful cDNA conversion and results in lowered signal amount (*see* Supplementary Note 3 and Fig. 2 in [13]). Primer can be positioned as close as 5 bp downstream SNP as long as PLP hybridization site is not overlapping with LNA bases (LNA bases likely prevent efficient PLP displacement during RCA).
4. ProbeMaker is available at <http://probemaker.sourceforge.net/> for both OSX and Windows platforms. Program allows importing single or batch cDNA targets in FASTA format for automated PLP design for each allele-specific variant (SNP within the sequence should be annotated as [N/N]). It is up to the user to define PLP arms binding properties like T_m and backbone sequence (including linkers and decorator probe motif). Since ProbeMaker does not correct the PLP arms T_m for the formamide presence so we typically set PLP arms T_m at 55–60 $^{\circ}$ C. Guide with examples how to use the software is available online.

5. Asymmetric arms are acceptable. We discourage designing overextended PLP arms. Even though the PLP can still generate detectable signal, high T_m can prevent probe melting if a PLP for another allele is transiently bound (in such case, the ligation will not take place and SNP will be “lost”). Monovalent and divalent ion concentrations, formamide, PLP concentration, and temperature influence the T_m of PLP arms. The hybridization and ligation reaction conditions exemplified in this chapter are the following: 75 mM monovalent ions, 10 mM divalent ions, 20% formamide, 0.1 μ M padlock probe concentration, and 45 °C.
6. PLP backbone consists of “decorator motif” and linkers (Fig. 2c). Decorator motif in the PLP backbone and the corresponding decorator oligonucleotide have identical sequences (20–25 bp). While decorator motifs are unique for every mRNA or SNP target, linkers can be propagated as long as PLP secondary structure is not compromised. We typically use series of adenine nucleotides to match correct backbone length as they have minimal effect on PLP secondary structure (other bases are acceptable but can have greater impact on the structure).
7. PLP phosphorylation protocol: 10 μ M final PLP concentration (can be changed if desired); 0.2 U/ μ L of T4 PNK kinase (e.g., Thermo Scientific); 1 $\times$ PNK kinase buffer; 1 mM ATP and H₂O in a final volume of 50 μ L. Mix should be incubated at 37 °C for 30 min, followed by enzyme inactivation at 65 °C for 20 min. Phosphorylated PLPs are stored at –20 °C for few months. Though we use large excess of primers and PLPs in both RT and ligation step of this protocol, slow retention of oligonucleotides on the wall of the tubes, combined with frequent freezing–thawing can influence probes performance. We recommend aliquoting oligonucleotides older than 6 months.
8. Decorator oligonucleotides can be conjugated with a fluorophore on any terminus. We routinely use 6-Carboxyfluorescein, Texas Red, Cyanine or Alexa Fluor[®] dyes. Decorator probe sequence should be blasted (<http://blast.ncbi.nlm.nih.gov/Blast.cgi>) against refseq mRNA database to minimize nonspecific oligo binding and autofluorescence build-up.
9. Though PLPs are surprisingly tolerant to secondary structures, one should select PLP with the highest free energy. Small internal loops or hairpins are acceptable but should be avoided in the region involving recognition of the target sequence (can limit hybridization of PLP arms) or within decoration motif (can hinder hybridization of the decorator probe). Shortening

an arm where a strong hairpin is present can aid hybridization and can be compensated by extending another arm. Finally, if PLPs are used in a set, formation of heterodimers or probability of probe-on-probe ligation should be evaluated. We commonly use “two-state melting” engine to evaluate multiple probe hybridization probability (<http://unafold.rna.albany.edu/?q=DINAMelt/Two-state-melting>).

10. Enzymes (including reverse transcriptase, RNase inhibitor, phi29 polymerase and RnaseH) from well-known vendors like NEB or Fermentas (Thermo Fisher Scientific) performs equally well in our hands.
11. DEPC inhibits RNases present in water, buffers or labware by irreversible, covalent modification of selected amino acids [21]. Following DEPC treatment, all solutions should be autoclaved to break down DEPC residue. Less dangerous alternatives to DEPC, such as DMPC, can be considered.
12. We recommend using freshly prepared formaldehyde solutions in DEPC-PBS. Working solutions can be prepared from either 37% methanol-stabilized stock solution or from paraformaldehyde powder. Aliquots of 3.7% formaldehyde in DEPC-PBS in 1 mL (used during the experiment) and 15 mL (for cell fixation) can be stored at -20°C . Do not freeze and thaw.
13. Lyophilized pepsin batches may vary in activity, even from the same supplier. Every new batch of pepsin should be tested on established model. We typically detect housekeeping gene (*ACTB* or *GAPDH*) in the same tissue type to evaluate detection reproducibility.
14. Far-red dyes are more susceptible to photobleaching. Slow-Fade[®] Gold Antifade Mountant works best in our experience.
15. Keep 0.1 v/v % DEPC in PBS or ddH₂O for at least 1 h at 37°C (or overnight at RT), followed by autoclaving.
16. Secure-Seal chambers come in different sizes, shapes and depths. For experiments performed on fixed cell lines, we typically use $\sim 50\ \mu\text{L}$ chambers (round, 9 mm diameter, and 0.8 mm deep). For larger cell areas or larger tissue specimen, 100 μL or 350 μL chambers can be used.
17. To achieve optimal optical resolution, cover glass thickness needs to be adjusted for the microscope setup used.
18. For low copy number transcripts we typically genotype *KRAS* codon 12 SNPs [13] and for abundant mRNAs, *ACTB* in human/mouse fibroblasts can be detected.
19. Unlike in many in situ hybridization methods, where cells are grown on coverslips, we grow cells on the microscope slides directly. We found slides to be more resistant to breaking (especially when applying and peeling the silicone chambers

off) and on the slides, multiple experiments can be run in parallel (up to eight), making the work more convenient. Also, multiple consecutive tissue section can be placed on a single slide to facilitate fast experimenting and imaging.

20. Optimal seeding conditions should be identified experimentally for every cell line. For cells with large cytoplasm, 3 mL suspension is usually enough to create homogenous cell layer on each slide. Cells with smaller cytoplasm can be seeded at higher density. In our experience, overnight incubation allows cells to adhere to slides efficiently. Extended incubation can result in cell proliferation on-slide (too dense or clumped cells are difficult to analyze by image analysis software).
21. Formaldehyde, HCl and formamide should be disposed in accordance to local lab regulations.
22. Tween 20, as a surfactant, coats the chambers, facilitates buffer exchange and prevents formation of "dead spaces" inside the chamber. As a detergent, it can provoke bubble formation. We recommend adding a buffer into a chamber when slide is slightly tilted (gravity helps to fill up the chamber evenly).
23. Experiment can be halted at this point if necessary. In such case, replace $1 \times \text{PBS-T}$ with $1 \times \text{PBS}$ and keep the slide at 4°C for up to 24 hours. We recommend proceeding with the protocol without interruptions until the cDNA is synthesized to minimize risk for mRNA degradation.
24. Any slides that enhance adhesion of tissue sections can be used. (SuperFrost Plus[®] from Menzel-Gläser work very well in our hands). Sections should be as thin as possible (preferably few cell layer). Thinner sections are more prone to break and fold during cutting. We commonly use $10\ \mu\text{m}$ thick sections.
25. The fixation time needs to compromise optimal fixative diffusion and minimize loss of tissue content. We recommend pragmatic evaluation of fixation parameters (consecutive sections should be used for each condition). Fixation time may vary for different tissues and different specimen thicknesses. Housekeeping gene is a good candidate for such optimization studies. Use conditions showing maximal signal amount.
26. RNase H has the highest activity at 37°C . It degrades RNA from mRNA/cDNA heteroduplex during the first 37°C incubation. After 30 min, sample is transferred to 45°C which is the optimal temperature for the *Tth* ligase. Addition of formamide into the mix lowers dsDNA stability (T_m of PLP arms/cDNA duplex). This enables extension of PLP arms that strengthens probe "locking" on cDNA and gives a good balance between arms melting and specific binding.

27. The optimal temperature for phi29 polymerase is 37 °C. We typically conduct RCA for 1 h. If RCA is performed for several hours (or over-night) at 37 °C, RCPs may start to fragment what could interfere with accurate signal counting. If large RCPs are desired (thick tissue sections or those with high autofluorescence), we advise doing RCA at RT over-night. Such approach will generate very large but compact RCPs.
28. In a multiplexed reaction (when more than one decorator probe is used), we recommend incubation at 37 °C for 30 min (to minimize nonspecific binding of the oligonucleotides). In such case, cover the chamber inlets to prevent mix evaporation.
29. A double edge razor can be used to facilitate complete removal of the Secure-Seal chamber.
30. We typically apply 5–7 µL of mounting medium for single, 50 µL Secure-Seal chamber. Remove excess of the medium by gently pressing the slide against a coverslip (excess of medium will be absorbed by the paper towel).
31. We use a Zeiss Axioplan II Epifluorescence microscope equipped with a 100-W mercury lamp and a Hamamatsu, C4742-95 CCD camera. The following filter setup provides good wavelength separation and minimal crosstalk between different channels. 38HE (Zeiss) for imaging GFP/FITC/FAM dyes; SP102v2 (Chroma) for imaging Cy3 (minimal cross talk with 38HE filter); SP103v2 (Chroma) for imaging Cy3.5/TexasRed; SP104v2 (Chroma) for imaging Cy5; 49007 (Chroma) for imaging Cy7/Alexa 7.5 dyes.
32. CellProfiler is a great, user-friendly tool to aid biologists in image processing and analysis. With respect to presented protocol, CellProfiler offers scripts for cell segmentation (definition of the nucleus and the cytoplasm), RCP segmentation, and assignment of RCPs to individual cells or fluorescence measurements. All scripts can be implemented in automated pipeline, allowing for batch image processing. An example script for cell and RCP identification is available at CellProfiler website <http://www.cellprofiler.org>. Briefly, gray scale TIFF images (offering highest resolution, JPEG images are processed faster and can also be used) from individual fluorescence channels are loaded into the pipeline. Cells are segmented to nuclei and cytoplasm and RCPs are identified and related to neighboring cells. Finally, number of RCPs for each cell is exported as a .csv file, which can be used for post-analysis processing.
33. The overall sensitivity of mRNA detection in situ in cell lines using the protocol provided has been estimated to be ~30% [13]. This efficiency parameter is valid for high quality samples

(freshly prepared cell lines) and might be lower in fresh frozen material or older, more degraded FFPE sections. mRNA detection efficiency can be further decreased if more than five LNA primers are used simultaneously (most likely due to formation of strong primer–primer hybrids, enhanced by the presence of LNA analogs). Pragmatic primer evaluation to minimize probability of primer dimer formation in the whole probe pool could possibly decrease this effect. Assuming that protocol was executed as presented and all reagents were fresh, fully active (enzymes) and RNase free, we still observe signal variation for certain mRNAs with similar RPKM values. If mRNA is highly structured in the target SNP region, redesigning an LNA primer (or using random decamers) can aid RT. Relatively low sensitivity of the in situ protocol can limit single cell mRNA detection for low expression level transcripts. Though we routinely detect single cell SNP in mRNAs with RPKM ~9 like *KRAS* ([22] Fig. S1, [23] Supplementary Table 1) successful qualitative or quantitative analyses of mRNA might not require single cell resolution. Measurement of *KRAS* codon 12 and 13 point mutations ratio in selected whole-specimen area was shown to be a good approach for molecular diagnostic scoring in colorectal cancer [22]. In [18], allele signal density estimation plots were used to elucidate expression pattern differences in specimen where single cell data was limited.

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